Freshwater Ecology & Aquatic Science

Multiple Choice Assessment – 100 Questions

Name: _____
Class/Period: _____ Date: _____

Directions: Select the best answer for each question. Circle A, B, C, or D.

1. A food web differs from a food chain because it:
A. Shows only producers
B. Shows multiple feeding relationships
C. Includes only predators
D. Represents one energy pathway

2. Primary producers in freshwater ecosystems are mainly:
A. Fish
B. Zooplankton
C. Algae and aquatic plants
D. Decomposers

3. Energy transfer between trophic levels is typically:
A. 100% efficient
B. 50% efficient
C. About 10% efficient
D. Completely lost

4. A population is defined as:
A. All organisms in an ecosystem
B. One species living in an area
C. Multiple species interacting
D. Abiotic factors in a habitat

5. Carrying capacity refers to:
A. Maximum population size an environment can support
B. Birth rate of a population
C. Immigration rate
D. Predator population size

6. Exponential population growth occurs when:
A. Resources are limited
B. Birth and death rates are equal
C. Resources are abundant
D. Predators increase

7. Logistic growth differs from exponential growth because it:
A. Has no limiting factors
B. Levels off at carrying capacity

C. Occurs only in aquatic systems
D. Has a constant growth rate

8. Competition occurs when organisms:
A. Share mutual benefits
B. Fight physically
C. Require the same limited resources
D. Eat each other

9. Predation is best described as:
A. A mutualistic interaction
B. One organism killing another for food
C. Both species benefiting
D. Two species harmed

10. Mutualism is a relationship where:
A. One benefits, one harmed
B. Both benefit
C. Both harmed
D. One unaffected

11. Commensalism occurs when:
A. Both species benefit
B. One benefits, the other is unaffected
C. One benefits, the other is harmed
D. Both species compete

12. Decomposers are important because they:
A. Increase oxygen levels
B. Recycle nutrients
C. Reduce primary productivity
D. Consume producers only

13. The water cycle includes all of the following EXCEPT:
A. Evaporation
B. Precipitation
C. Transpiration
D. Photosynthesis

14. Runoff most directly affects freshwater systems by:
A. Increasing salinity
B. Transporting nutrients and pollutants
C. Reducing sunlight
D. Decreasing water volume

15. Eutrophication is caused primarily by excess:
A. Oxygen
B. Nitrogen and phosphorus
C. Carbon dioxide
D. Salinity

16. Algal blooms can lead to:

A. Increased dissolved oxygen
B. Fish kills
C. Reduced nutrient levels
D. Lower temperatures

17. Which organism is a secondary consumer?
A. Algae
B. Zooplankton
C. Small fish eating zooplankton
D. Aquatic plants

18. A keystone species:
A. Has the largest population
B. Has a disproportionately large effect on its ecosystem
C. Is always a predator
D. Is a producer

19. Biodiversity generally increases ecosystem:
A. Instability
B. Vulnerability
C. Productivity and resilience
D. Pollution levels

20. Trophic cascades occur when:
A. Nutrients cycle through soil
B. Top predators affect lower trophic levels
C. Producers increase rapidly
D. Decomposers disappear

21. Limiting factors can be:
A. Only biotic
B. Only abiotic
C. Biotic and abiotic
D. Neither

22. Immigration increases population size by:
A. Births
B. Deaths
C. Individuals entering a population
D. Individuals leaving a population

23. Emigration causes population size to:
A. Increase
B. Remain constant
C. Decrease
D. Double

24. A community includes:
A. One species
B. All abiotic factors
C. All interacting populations
D. Only producers

25. Succession in freshwater ecosystems often begins with:
A. Fish
B. Large plants
C. Algae and microorganisms
D. Mammals

26. pH measures:
A. Temperature
B. Salinity
C. Acidity or alkalinity
D. Turbidity

27. A pH of 7 is considered:
A. Acidic
B. Neutral
C. Basic
D. Toxic

28. Most freshwater organisms prefer pH values between:
A. 2–4
B. 5–9
C. 10–12
D. 12–14

29. Dissolved oxygen (DO) is essential for:
A. Photosynthesis
B. Respiration in aquatic organisms
C. Nutrient cycling
D. Sedimentation

30. Cold water generally holds:
A. Less oxygen than warm water
B. More oxygen than warm water
C. No oxygen
D. Equal oxygen

31. Salinity refers to:
A. Amount of sediment
B. Concentration of dissolved salts
C. Water clarity
D. Oxygen content

32. Freshwater is defined as having salinity less than:
A. 35 ppt
B. 10 ppt
C. 5 ppt
D. 0.5 ppt

33. High nitrate levels often indicate:
A. Acid rain
B. Agricultural runoff

C. Thermal pollution
D. Natural weathering

34. Phosphates are important because they:
A. Increase salinity
B. Promote plant and algal growth
C. Reduce oxygen directly
D. Lower pH

35. Biochemical Oxygen Demand (BOD) measures:
A. Oxygen produced by algae
B. Oxygen required to decompose organic matter
C. Total dissolved solids
D. Water temperature

36. High BOD usually results in:
A. High dissolved oxygen
B. Low dissolved oxygen
C. Increased pH
D. Increased clarity

37. Total solids include:
A. Only dissolved substances
B. Only suspended substances
C. Dissolved and suspended materials
D. Only organic matter

38. Turbidity affects aquatic life by:
A. Increasing photosynthesis
B. Blocking light penetration
C. Increasing oxygen levels
D. Raising salinity

39. Which factor most directly affects fish respiration?
A. pH
B. Dissolved oxygen
C. Nitrates
D. Salinity

40. Thermal pollution usually:
A. Increases dissolved oxygen
B. Decreases dissolved oxygen
C. Lowers nutrient levels
D. Improves water quality

41. Acid rain primarily affects water by:
A. Increasing temperature
B. Lowering pH
C. Raising salinity
D. Increasing turbidity

42. A sudden drop in DO could result from:

A. Algal die-off
B. Cold temperatures
C. Clear water
D. Low BOD

43. Which chemical parameter is measured in parts per million (ppm)?
A. Temperature
B. Dissolved oxygen
C. pH
D. Flow rate

44. Excess nutrients often lead to:
A. Oligotrophic conditions
B. Eutrophic conditions
C. Reduced productivity
D. Increased biodiversity

45. Chemical testing is important because it:
A. Identifies species
B. Predicts weather
C. Indicates water quality
D. Measures population size

46. Potable water is:
A. Wastewater
B. Drinkable water
C. Saltwater
D. Untreated water

47. The first step in most water treatment plants is:
A. Chlorination
B. Filtration
C. Coagulation and sedimentation
D. Fluoridation

48. Sedimentation removes:
A. Dissolved chemicals
B. Microorganisms
C. Large suspended particles
D. Salts

49. Filtration primarily removes:
A. Gases
B. Large debris
C. Fine particles
D. Dissolved ions

50. Chlorination is used to:
A. Remove solids
B. Kill pathogens
C. Add nutrients
D. Increase turbidity

51. Wastewater treatment protects:
A. Only humans
B. Only fish
C. Both ecosystems and human health
D. Industry only

52. Primary wastewater treatment focuses on:
A. Biological processes
B. Chemical disinfection
C. Physical removal of solids
D. Nutrient removal

53. Secondary treatment uses:
A. Physical screens
B. Microorganisms
C. Chlorine
D. Sedimentation tanks

54. Tertiary treatment often removes:
A. Large debris
B. Pathogens only
C. Excess nutrients
D. Oil

55. A watershed is:
A. A river
B. A lake
C. Land draining into a common water body
D. A groundwater aquifer

56. Activities upstream can:
A. Never affect downstream water
B. Affect downstream water quality
C. Only affect groundwater
D. Improve water quality automatically

57. Sedimentation pollution mainly comes from:
A. Sewage
B. Soil erosion
C. Acid rain
D. Industrial discharge

58. Excess sediment can:
A. Improve fish habitat
B. Smother benthic organisms
C. Increase DO
D. Reduce water temperature

59. Best management practices (BMPs) are designed to:
A. Increase pollution
B. Reduce runoff and erosion

C. Increase fertilizer use
D. Eliminate watersheds

60. Riparian buffers help by:
A. Increasing salinity
B. Filtering runoff
C. Removing oxygen
D. Heating water

61. Point source pollution is:
A. Diffuse runoff
B. From a single identifiable source
C. Always natural
D. Unregulated

62. Nonpoint source pollution includes:
A. Factory discharge pipes
B. Sewage outlets
C. Agricultural runoff
D. Wastewater plants

63. Sustainable water management focuses on:
A. Short-term use
B. Long-term availability and quality
C. Maximizing extraction
D. Ignoring ecosystems

64. Overuse of groundwater can cause:
A. Increased rainfall
B. Aquifer depletion
C. Improved water quality
D. Higher salinity in rivers

65. Water conservation helps by:
A. Increasing pollution
B. Reducing demand
C. Lowering biodiversity
D. Increasing runoff

66. Macroinvertebrates are organisms that are:
A. Microscopic
B. Vertebrates
C. Visible to the naked eye and lack backbones
D. Plants

67. Which is a pollution-sensitive macroinvertebrate?
A. Leech
B. Midge larva
C. Mayfly nymph
D. Aquatic worm

68. Which organism indicates poor water quality?

A. Stonefly
B. Caddisfly
C. Mayfly
D. Bloodworm

69. Macroinvertebrates are useful indicators because they:
A. Move frequently
B. Have short life spans
C. Reflect long-term water conditions
D. Are easy to remove

70. Shredders primarily feed on:
A. Algae
B. Leaf litter
C. Other insects
D. Fish

71. Scrapers feed mainly on:
A. Detritus
B. Periphyton and algae
C. Leaves
D. Zooplankton

72. Predatory macroinvertebrates eat:
A. Plants
B. Detritus
C. Other animals
D. Algae

73. Dragonfly nymphs are:
A. Pollution-sensitive
B. Filter feeders
C. Predators
D. Shredders

74. Caddisfly larvae are known for:
A. Burrowing
B. Building protective cases
C. Parasitism
D. Filter feeding only

75. Life cycles of aquatic insects typically include:
A. Egg, nymph, adult
B. Egg, larva, pupa, adult
C. Egg, tadpole, adult
D. Larva only

76. Invasive species are harmful because they:
A. Increase biodiversity
B. Have no predators
C. Outcompete native species
D. Improve water quality

77. Zebra mussels cause problems by:
A. Increasing turbidity
B. Clogging intake pipes
C. Reducing filtration
D. Decreasing clarity

78. Asian carp are problematic because they:
A. Are predators of fish
B. Outcompete native species for food
C. Increase oxygen levels
D. Improve habitats

79. Macroinvertebrate diversity generally indicates:
A. Poor water quality
B. Stable water chemistry
C. High pollution
D. Low oxygen

80. Tolerant species can survive:
A. Only pristine water
B. Moderate to poor water quality
C. High oxygen only
D. Cold temperatures only

81. A biotic index is based on:
A. Chemical tests
B. Species tolerance values
C. Water temperature
D. pH only

82. Which group is MOST sensitive to pollution?
A. Aquatic worms
B. Leeches
C. Stoneflies
D. Midges

83. Nuisance species often spread due to:
A. Natural predators
B. Human activity
C. Improved water quality
D. Low nutrient levels

84. Macroinvertebrates are sampled because they are:
A. Easy to culture
B. Stationary and abundant
C. Large predators
D. Short-lived

85. A decline in sensitive species usually indicates:
A. Improved water quality
B. Increased pollution

C. Stable conditions
D. Higher biodiversity

86. Water monitoring helps scientists:
A. Predict earthquakes
B. Assess ecosystem health
C. Identify all species
D. Control weather

87. A secchi disk measures:
A. pH
B. Turbidity
C. Dissolved oxygen
D. Salinity

88. Stream flow is measured using:
A. pH meters
B. Flow meters
C. Secchi disks
D. Thermometers

89. Repeated sampling over time allows scientists to:
A. Eliminate error
B. Identify trends
C. Remove pollution
D. Increase nutrients

90. A sudden change in data may indicate:
A. Equipment error or pollution event
B. Stable conditions
C. Natural balance
D. No significance

91. Dissolved oxygen is often lowest:
A. At dawn
B. At noon
C. In winter
D. After rainfall

92. Macroinvertebrate surveys are considered:
A. Chemical monitoring
B. Physical monitoring
C. Biological monitoring
D. Thermal monitoring

93. Replication in experiments increases:
A. Bias
B. Accuracy and reliability
C. Cost only
D. Error

94. A control site is used to:

A. Add pollution
B. Compare natural conditions
C. Replace experimental data
D. Eliminate variables

95. Data tables are useful because they:
A. Replace graphs
B. Show raw values clearly
C. Eliminate analysis
D. Reduce accuracy

96. Graphs help by:
A. Hiding trends
B. Visualizing patterns
C. Increasing error
D. Replacing sampling

97. Which unit is commonly used for DO?
A. ppt
B. ppm or mg/L
C. °C
D. NTU

98. Monitoring watersheds helps manage:
A. Air quality
B. Soil only
C. Water resources
D. Wildlife only

99. Citizen science programs are valuable because they:
A. Replace professionals
B. Increase data collection
C. Reduce accuracy
D. Eliminate costs

100. Interpreting data requires:
A. Guessing
B. Comparing results to standards
C. Ignoring variability
D. Using one measurement